Malhar Gudekar

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Education

University of Illinois Urbana Champaign August 2024 - May 2026
Master of Science, Information Management GPA: 3.8/4
Coursework: Data Warehousing & Business Intelligence, Information Management, Information Consulting, Methods of Data Science

University of Mumbai June 2019 - May 2023
Bachelor of Engineering, Computer Science GPA: 9.25/10
Coursework: DBMS, Big Data Analysis, Data Warehousing and Mining, Project Management, NLP, Machine Learning, Social Media Analytics

Skills

Quantitative & Statistical: Time Series Analysis, Statistical Modeling, Predictive Analytics, Feature Engineering, Backtesting, ARIMA, Regression, Classification, Model Evaluation, Hypothesis Testing, Monte Carlo Simulation
Programming & OOP: Python (OOP, NumPy, Pandas, Scikit-learn, LightGBM), SQL, R, C++ (basic)
Systems & Data: PostgreSQL, Linux, ETL Pipelines, FastAPI, Kafka, PySpark, AWS, Docker, Git

Work Experience

University of Illinois Urbana Champaign May 2025 - Present
Research Assistant - iSchool
- Designed and implemented end-to-end **statistical modeling pipeline** in Python to ingest and analyze time-series wearable sensor data from 100+ users, applying feature engineering and temporal pattern extraction to surface longitudinal behavioral signals for downstream predictive analytics
- Optimized PostgreSQL schema, indexing strategies, and query execution plans for high-volume time-series datasets, improving query throughput by **38%** and enabling low-latency access to model-ready features in production Linux environments
- Built embedding-based retrieval and summarization pipeline to enhance contextual understanding of user interactions, improving multi-turn response accuracy by 43% and supporting more reliable behavioral analysis

Research Assistant - CHI AI in Infodemic Management January 2025 - May 2025
- Conducted full statistical research cycle in Python — dataset evaluation, feature selection, model prototyping, and performance monitoring — on large-scale social media corpora, reducing inference latency by **41%** and improving classification precision through iterative model tuning
- Built distributed **data pipelines** using **Kafka** and **Neo4j** Streams to enable **scalable ingestion**, transformation, and real-time analysis of high-volume unstructured datasets
- Designed production-grade REST APIs in FastAPI to serve trained statistical models, implementing input validation, performance monitoring, and automated deployment workflows ensuring reproducible, low-latency inference across distributed Linux infrastructure

Business Intelligence Group August 2025 - December 2025
Technical Consultant
- Designed and deployed end-to-end **quantitative data pipeline** combining ETL, feature engineering, and vector-based retrieval to improve document scoring precision by **64%**, maintaining fault-tolerant operation under production load
- Designed scalable **ETL** and feature pipelines using **FastAPI** and **PostgreSQL** to transform unstructured data into structured formats suitable for analytics and machine learning use cases
- Profiled and optimized indexing strategies and query execution plans to achieve low-latency model inference, reducing system response time by **64%** and enabling real-time analytical serving in a production environment

Swift Mobil Software Solutions Provider July 2023 - October 2023
Data Analyst
- Analyzed large-scale operational datasets using **Python** and **PySpark** to identify trends, anomalies, and performance issues, combining exploratory analysis and root cause investigation to support data-driven decision-making
- Built **Power BI** dashboards to track key performance indicators and improve visibility into operational performance, reducing reporting time by 40%

Pscope Technologies Pvt. Ltd. January 2023 - June 2023
Data Analyst
- Developed and maintained Power BI dashboards using **SQL** and **DAX**, transforming raw data into actionable insights for business decision-making across multiple domains.
- Performed data validation, cleaning, and transformation on large multi-source datasets using **SQL** and **Excel**, improving data accuracy by over 30% and reducing reporting inconsistencies, enabling faster and more reliable decision-making across sales, finance, and operations teams

Stu/Dio at Illinois August 2025 - Present
Project Manager
- Led a **cross-functional** team of six to deliver and scale Deepcover game enhancements by leveraging user data, performance metrics, and feedback analysis to drive feature prioritization and improve games outcomes

Projects

Scalable ML System for Customer Risk Prediction
- Built end-to-end quantitative modeling system in Python using Pandas and LightGBM to forecast customer spending from large-scale **time-series transaction data**, applying feature engineering on behavioral trends and temporal patterns to improve prediction precision by **20%**
- Designed and implemented **backtesting framework** to validate model performance across rolling historical time windows, stress-testing predictions under distribution shift and varying market conditions
- Developed OOP-structured decision framework for risk-aware credit line classification, encapsulating feature pipelines, model inference, and threshold logic in a maintainable, modular Python codebase